A Mellin-Barnes closed form for a pass sum in Kim and Thorne (1991)

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What I am asking. Two things: does it hold, and is it already known? The full proofs are in the companion write-up in the same repository, "Proof pieces for a Mellin-Barnes closed form of a pass sum" (version 1.1, 50 pages, revised October 29, 2026).

1. The object

Fix real parameters 0 < ξ < 1 and r > 0, and let u > 0. For each integer N ≥ 1 put

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w_N = ξ^N,   a_N = r^N (1 + 4 w_N + w_N^2),   b_N = ξ^(-N) - 1 (so b_N > 0),   h(y) = (1 + y)^(-3) for y > -1,

Ψ(u) = sum over N ≥ 1 of a_N h(u b_N).
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Write L = ln(1/ξ) > 0 and p = ln(r)/L, so r = ξ^(-p). The number p may be any real number; p < 0 exactly when r < 1. For complex z and an integer j ≥ 0, C(z, j) = z (z - 1) ... (z - j + 1)/j!.

Convergence. For u > 0 the N-th term equals (r ξ^3)^N (1 + 4 w_N + w_N^2)/(u + (1 - u) w_N)^3, which behaves like (r ξ^3)^N/u^3 as N grows. The sum therefore converges exactly when r ξ^3 < 1, that is, when p < 3.

2. Where it comes from

Kim and Thorne, Phys. Rev. D 43, 3929 (1991), eq. 67, write a renormalized stress-energy tensor as a sum over N of terms of this shape and leave the sum unevaluated. Ψ is that sum after a change of variables. The physical reading of the model belongs to its authors; nothing here rests on it or claims it.

In the paper's symbols, r = (b/2D) ξ^(-2) and u = dT/c, with dT the coordinate time left before the Cauchy horizon of their eq. 63 and c = ρ^2 D/(2 X (D - X)); a prefactor in D, X and c stands outside the sum. This is algebra on the printed eq. 67.

3. The representation

H(s) = integral from 0 to infinity of y^(s-1) h(y) dy = Γ(s) Γ(3 - s)/2 = π (1 - s)(2 - s)/(2 sin(π s)), for 0 < Re s < 3 (a Beta integral). Φ(s) = sum over N ≥ 1 of a_N b_N^(-s), absolutely convergent for Re s > p. For p < 3, u > 0 and any c with max(0, p) < c < 3:

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Ψ(u) = (1/(2 π i)) times the integral along Re s = c of H(s) Φ(s) u^(-s) ds.
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Φ continues to a meromorphic function on the whole plane, Φ(s) = sum over n ≥ 0 of Q_n(s) ρ_n(s), with ρ_n(s) = 1/(e^(L(n + s - p)) - 1) and Q_n(s) = sum over m = 0 to min(2, n) of c_m C(s + n - m - 1, n - m), where (c_0, c_1, c_2) = (1, 4, 1). Its poles are simple, at s_{n,k} = p - n - 2 π i k/L (n = 0, 1, 2, ...; k any integer), with residue Q_n(s_{n,k})/L (a zero residue means no pole there).

4. What "closed form" means here (claims C1 to C4)

Not a finite formula in elementary functions. It means the exact integral of section 3, valid for every u > 0, together with two residue expansions that are convergent series of explicit terms (Gamma-type factors, the polynomials Q_n, and finite sums of geometric sums).

- **C1, large u.** For u > ξ/(1 - ξ): Ψ(u) = sum over l ≥ 0 of (-1)^l C(l + 2, 2) u^(-(3 + l)) Φ(3 + l). The series converges there and diverges for 0 < u < ξ/(1 - ξ), so ξ/(1 - ξ) is its exact radius.
- **C2, small u, p not an integer.** For 0 < u < 1: Ψ(u) = sum over l ≥ 0 of (-1)^l C(l + 2, 2) Φ(-l) u^l + (1/L) sum over n ≥ 0 and all integers k of H(s_{n,k}) Q_n(s_{n,k}) u^(-s_{n,k}), both series absolutely convergent, each Φ(-l) a finite sum of geometric sums. The l-series diverges for u > 1, so 1 is its radius.
- **C3, integer p.** When p is an integer, the pole of H at s = -l and the pole of Φ at s = p - n, with n = l + p, meet. Their two residues, each growing without bound as p approaches the integer, combine into one finite term that tends to a logarithm. At the integer the pole is double and gives a term F u^l plus a term G u^l ln(1/u). Section 5 says exactly where G is zero.
- **C4, the ripple.** The terms with k not zero come in conjugate pairs, u^(-(p - n)) times u^(+2 π i k/L) and its conjugate; each pair sums to a cosine in ln u with period ln(1/ξ)/k. The poles are spaced 2 π/ln(1/ξ) apart in Im s; the ripple's period in ln u is ln(1/ξ). An earlier draft gave 2 π/ln(1/ξ); section 6 reports a test of that wrong value.

The gap. For ξ > 1/2, ξ/(1 - ξ) > 1, so for 1 ≤ u ≤ ξ/(1 - ξ) neither series converges and the integral of section 3 is the closed form there.

5. Where the logarithms vanish: a Pell chain

At an integer p and l ≥ max(0, -p), the coefficient of u^l ln(1/u) in C3 is G_l = R_l Q_{p+l}(-l)/L with R_l = (-1)^l C(l + 2, 2), never 0, so the logarithm is absent exactly when Q_{p+l}(-l) = 0. The companion write-up's Proposition 4.4 says where:

- If p ≥ 3, every Q_{p+l}(-l) is 0 (and Φ is regular at every s = -l).
- If p = 2, Q_{p+l}(-l) = (-1)^l, never 0.
- If p ≤ 1, put b = 5 - 2p. Then Q_{p+l}(-l) = 0 exactly when 3b^2 - 2 = y^2 for an integer y ≥ 0 and l = (3b - 6 ± y)/2 (the lower sign only when that is at least 0).

So the integers p ≤ 1 where a logarithm drops out are exactly those with y^2 - 3b^2 = -2, that is y + b√3 = (1 + √3)(2 + √3)^k:

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p = 1, -3, -18, -74, -283, -1063, -3974, ...      p_{k+1} = 4 p_k - p_{k-1} - 5
vanishing l = 4, 23, 94, 359, 1348, 5039, ...       l_{k+1} = 4 l_k - l_{k-1} + 6
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Each vanishing l belongs to two neighbouring p's: for example, u^4 ln(1/u) is absent at p = 1, and u^4 ln(1/u) and u^23 ln(1/u) at p = -3. The (y, b) pairs on the chain are (5, 3), (19, 11), (71, 41), (265, 153), (989, 571), ...; the fourth is Archimedes' lower bound 265/153 < √3 (Measurement of a Circle, Proposition 3). That a classical bound sits on this chain is a fact about the equation y^2 - 3b^2 = -2; nothing classical is claimed as new. What is specific here is that these meeting residues vanish exactly on pairs of consecutive solutions.

Checked in exact integer arithmetic: for every integer p from 2 down to -200,000 the discriminant is a square exactly on the chain, y^2 - 3b^2 = -2 has no other positive solution with b ≤ 400,005, and the coefficient G_l agrees with an independent numerical route at 48 meeting pairs, the vanishing pairs (-3, 4) and (1, 4) among them.

6. Standing, at its real size

Written proofs exist in the companion write-up: Lemma L, an exact representation of Φ on Re s < 0 as an absolutely convergent double series of continued Beta functions; a growth bound for Φ far to the left at every real p (Theorem P3); the expansion of Ψ(u) for 0 < u < 1 at every p < 3, integer p included (Theorem P1); what converges for u ≥ 1 when p is not an integer (Theorem P2, its part (e) proved under a Diophantine condition and only outlined without it); and Proposition 4.4 of section 5. Every proof was checked by an AI instrument and read by no mathematician. "Theorem" there is only the label of a numbered statement with its proof attached.

Open: a mathematician's read; a fuller literature search; the unconditional form of Theorem P2 (e); Theorem P2 at integer p; certified (interval) arithmetic for the floating-point checks.

Evidence. Four pre-registered blind runs, rules and bars frozen before the points were drawn, 12 cases a run and nine points a case. 432 of 432 points matched the direct sum: largest relative difference 1.57 x 10^-15, smallest 1.46 x 10^-56, bar 10^-8. A separate sign rule in the same runs read one case wrong. A point is one (ξ, p, u); a case is one (ξ, p); the relative difference is |series - direct sum|/|direct sum|. All 432 of those points sat on the small-u side, u between 10^-4 and 10^-2 times ξ/(1 - ξ), with ξ from 1/35 to 17/25.

A fifth blind run (October 23, 2026) drew its points in the four regions the first four never reached: large u, u near 1, whole-number p (with p near a whole number), and p from 2.90 to 2.99, with ξ up to 0.943. The bars were frozen before the points were drawn, and the seed of the draw was committed by its hash before it was revealed. An independent high-precision direct sum, the referee, was good at all 123 points (117 of them scored). The closed-form routes (C1, C2, C3 and the integral of section 3) matched it at all 238 route readings against the bar of 10^-8; the largest relative difference was 5.8 x 10^-20 for the series routes and 3.4 x 10^-14 for the integral, and those are the limits of the runs' own arithmetic: a separate check with its own 30-digit code found the

closed form agreeing with the direct sum to better than 10^-24 at 27 of the points. Planted wrong formulas were caught wherever they were judged.

The period, tested. On the same points, a planted copy of the routes using the earlier draft's wrong period ($2\pi/\ln(1/\xi)$ in place of $\ln(1/\xi)$) was rejected at all 69 series points where it was tried, including the 54 where a one-percent error in the true period was too small for the bar to judge: the wrong period moves the ripple's poles toward the real axis, so the wrong ripple is damped less and shows everywhere. That test was not blind; it measures the bar against a planted error.

What the runs are not. Agreement at points is not a proof, and says nothing between the points or outside the ranges tested (ξ above 0.943, p above 2.99, u above 5,000 $\xi/(1 - \xi)$). The ripple is the part the bar sees least: where it is under about 10^-10 of the sum, a dropped ripple pair passes the bar, and in the fifth run it was judged at only 11 of 69 series points.

Novelty. Not found so far in the sources reached (the references below). The Mellin method itself is standard (Flajolet, Gourdon and Dumas, 1995). The case of Lemma L with no shift is Ueno and Nishizawa's (1995). Pell's equation is classical, and the b values 1, 3, 11, 41, 153, 571, ... are a known integer sequence (OEIS A001835). Whether the vanishing set of section 5 appears in published work has not been searched; that is one of the two questions I am asking.

7. Terms a reader may have found undefined, and where each is defined

- u and r: pure parameters here (section 1); their physical meaning is Kim and Thorne's (section 2 gives the change of variables).
- p: derived, $p = \ln r/\ln(1/\xi)$ (section 1). The two sides $p < 0$ and $0 < p < 3$ behave differently as u goes to 0.
- "closed form": section 4. "Two terms meet": C3. "Ripple" and its period: C4. The gap band for $\xi > 1/2$: end of section 4.
- "Pell chain", "vanishing l": section 5.
- "blind run", "bar", "relative difference", "point", "case": section 6.

Drafted with AI assistance and verified against my records. Reviewed and approved by Ron Hampton.

References: where this sum stands in the published literature

Citations 1 to 6 were matched at their DOI registry records or arXiv pages on October 23, 2026; 7 to 10 are as cited in the companion write-up.

1. **Kim, S.-W., and Thorne, K. S. (1991).** Do vacuum fluctuations prevent the creation of closed timelike curves? Physical Review D 43(12), 3929 to 3947. DOI 10.1103/PhysRevD.43.3929.
   The source. Eq. 67 writes the stress-energy tensor as a sum over N and leaves it unevaluated; the text says "we have computed only the leading-order part".
2. **Klinkhammer, G. (1992).** Vacuum polarization of scalar and spinor fields near closed null geodesics. Physical Review D 46(8), 3388 to 3394. DOI 10.1103/PhysRevD.46.3388.
   Takes the leading term and shows the series converges; does not sum it, and notes lower-order terms it does not compute (as read in his 1992 Caltech PhD thesis, chapter 3).
3. **Visser, M. (1994).** Hawking's chronology protection conjecture: singularity structure of the quantum stress-energy tensor. Nuclear Physics B 416, 895 to 906. arXiv:hep-th/9303023.
   Takes the leading term (its eq. 34); the sum over geodesics is left in place, with no closed form.
4. **Flajolet, P., Gourdon, X., and Dumas, P. (1995).** Mellin transforms and asymptotics: harmonic sums. Theoretical Computer Science 144(1-2), 3 to 58. DOI 10.1016/0304-3975(95)00002-E.
   The standard method for sums of this shape; it does not treat this sum.
5. **Paris, R. B., and Kaminski, D. (2001).** Asymptotics and Mellin-Barnes Integrals. Cambridge University Press. DOI 10.1017/CBO9780511546662.
   A book on the method.
6. **NIST Digital Library of Mathematical Functions, section 2.5, Mellin Transform Methods.** https://dlmf.nist.gov/2.5. Open access.
7. **Ueno, K., and Nishizawa, M. (1995).** Quantum groups and zeta-functions. In Quantum Groups (Karpacz, 1994), PWN, Warsaw, 115 to 126. arXiv:hep-th/9408143.
   Lemma L with no shift is their sum.
8. **Archimedes, Measurement of a Circle,** in T. L. Heath (ed.), The Works of Archimedes, Cambridge University Press, 1897, pages 91 to 98.
   Proposition 3 states 265/153 < √3 < 1351/780 without explanation (Heath's note, page 93).
9. **Davies, E. B. (2011).** Archimedes' calculations of square roots. arXiv:1101.0492.
   Lists the solutions of a^2 - 3b^2 = -2 that are the chain of section 5.
10. **The On-Line Encyclopedia of Integer Sequences,** A001835.